

HARIHARAN TS

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SUMMARY

Recent Computer Science (AI & ML) graduate with hands-on experience building production-grade ML pipelines, OCR-NLP preprocessing systems, and CNN-based classifiers. Shipped a full-stack AI finance application integrating multi-class expense classification, anomaly detection, and forecasting via FastAPI. Proficient in TensorFlow, Python, and REST API design. Actively building and experimenting with LLM-based applications, RAG systems, and AI agent workflows.

TECHNICAL SKILLS

Languages: Python (primary), Java, SQL

Backend: FastAPI, REST API Design, Async I/O, Request Validation, Middleware, Error Handling

Data & Storage: SQLite, Offline-first Architecture, Cloud Storage Integration, JSON Schema Design

AI Integration: TensorFlow, Scikit-learn, OCR (Tesseract), NLP Preprocessing, ML Inference as a Service

Frontend: React.js, Context API, Canvas API, Lazy Loading, Component Architecture

Core CS: Data Structures & Algorithms, OOP — 50+ problems solved in Java across arrays, trees, graphs & DP

Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook.

PROJECTS

AI-Powered Personal Finance Management System

2024 – Present

Python · FastAPI · TensorFlow · Scikit-learn · Tesseract OCR · NLP · SQLite · Cloud Storage

- Architected an end-to-end AI pipeline — data ingestion → OCR → NLP → classification → analytics — processing SMS, UPI alerts, and receipt images with optimized latency for real-time expense tracking
- Trained a multi-class expense classification model on 5,000+ transactions, achieving ~90% accuracy; implemented rule-based + ML hybrid logic to improve categorization reliability
- Built intelligent financial insights including budget alerts and spending summaries; integrated backend with FastAPI and Flutter frontend, supporting scalable data handling with SQLite and Firebase cloud sync

Traffic Sign Detection System

2024

Python · TensorFlow · CNN · OpenCV · Scikit-learn · GTSRB Dataset

- Trained 43-class CNN on 50,000+ GTSRB images achieving **96.4% test accuracy**; augmentation and class-weighted loss reduced misclassification by **22%**.
- Evaluated per-class precision/recall and optimised confusion-matrix hotspots via class-weighted loss; structured output for integration with ITS pipelines

Interactive One Piece Encyclopedia — Web App

2025 – Ongoing

React.js · JavaScript · Context API · Canvas API · JSON · Lazy Loading

- Designed a normalised JSON schema modelling 200+ characters, 30+ arcs, and geographic entities with relational references enabling O(1) cross-entity lookups
- Engineered a coordinate-based map rendering engine using Canvas API overlays to visualise world geography interactively across variable viewports
- Reduced initial bundle load time by ~**40%** via React. lazy, route-based code-splitting, and memoised component trees; preserved state across 15+ route transitions with Context API

EDUCATION

B.E. Computer Science & Engineering (AI & ML) — Sethu Institute of Technology, Madurai 2022 – 2026

CGPA: 8.79 · Honors: Full Stack Development

Class XII (Higher Secondary) Thiagarajar Model Higher Secondary School 2022 | 84%

Class X (SSLC) Thiagarajar Model Higher Secondary School 2020 | 83.4%

ACHIEVEMENTS & CERTIFICATIONS

- Department Topper — Academic Star & Above and Beyond honors · Winner in 7+ Technical symposiums · Top 5, Hackathon — Mohammed Sathak Engineering College
- AI Master Class (30-day) — Novitech Pvt Ltd · AI Primer — Infosys Springboard · 2× Robotics Training — Zetspire Technologies